

+ Looking Back: Catastrophic Forgetting



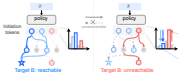
Old-task performance ↓

+ Looking Forward: Support Reshaping

Support view (initiation space)

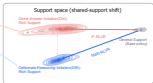
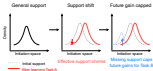
Before learning Task A

After learning Task A



Training on Task A reshapes the initiation support, making the target for verifying Task B unreachable.

Continual On-policy Optimization



Math-RLVR f_x

Deliberate reasoning

reasoning steps

Ans. $\hat{a}$

verified scope

IF-RLVR $\hat{a}$

Answer to follow user's constraints

verified scope

$\{x \in \text{scope}\} \checkmark$
 $\{x \notin \text{scope}\} \times$
 $\{x \in \text{scope}\} \checkmark$
 $\{x \notin \text{scope}\} \times$